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20251025	Mr. Richard Kho

I. BUSINESS UNDERSTANDING

Your task is to build a classification model that achieves the highest possible accuracy by experimenting with different combinations of data preparation techniques. You are expected to apply data sanitation, feature selection, and feature extraction methods to improve model performance and systematically compare their effects. As you move through the phases—from Business Understanding to Modeling—consider how each preprocessing choice influences the final outcome. Your goal is not just to build a model, but to demonstrate the value of thoughtful data preparation in predictive analytics. Moreover, your objective is to identify and explore three classification algorithms not currently included in your existing setup, then install them to extend WEKA's functionality.

Reflection (10 pts):

a. Provide details of the three (3) new classification algorithms, including their names and a brief description. (10 pts)

- The objective of this experiment is to build and evaluate a classification model that predicts the "Attack_Type" field based on the dataset's features. The goal is to achieve the highest possible accuracy by applying appropriate preprocessing and model selection techniques.

For this laboratory, I used NaiveBayes, DecisionTable, and J48 (C4.5 Decision Tree) because I can't use any other classification techniques because my WEKA keeps on erroring because it lacks MaxHeap. I used them to classify the dataset and compare their performance in WEKA. These algorithms were chosen because they represent three distinct approaches to classification — probabilistic, rule-based, and tree-based learning — which allows for a balanced comparison of interpretability and accuracy.

Through this process, I aimed to understand how different algorithms handle the same dataset and how data preparation affects predictive accuracy.

II. DATA UNDERSTANDING

Extract the dataset from our Git repository located in the “20251025-Lab-Wk12” directory, and perform your own assessment and evaluation of its contents. Please note that the class field is “Action Taken.”.

Reflection (40 pts):

1) Based on your assessment, what do you think the dataset is about? (5 pts)

2) What are the details of the dataset:

a) List the field names along with their corresponding data types. (15 pts)

- The dataset contains network or system activity records that indicate whether an event represents a specific type of cyber attack or a normal connection. Each row corresponds to an observation of traffic behavior with several features describing characteristics like duration, source, protocol, and packet details. The “Attack Type” column specifies the class or category of each record (e.g., DoS, Probe, Normal, etc.).

b) Indicate the number of records. (2 pts)

- The dataset includes multiple fields with different data types. Attributes such as duration, src_bytes, dst_bytes, wrong_fragment, and urgent are numeric, as they represent measurable quantities in the network traffic. Meanwhile, features like protocol_type, service, flag, and land are nominal, since they describe categorical properties such as the communication protocol or connection status. The “Attack Type” column, which serves as the class label, is also nominal because it categorizes each record into specific classes or labels that represent different attack types.

c) Identify the attribute in the dataset that is suitable to serve as the class label. (3 pts)

- The “Attack Type” column, which serves as the class label, is also nominal because it categorizes each record into specific classes or labels that represent different attack types.

3) Provide a thorough analysis of the dataset in terms of its readiness for model evaluation. Do you observe any anomalies in the dataset? (15 pts)

- Upon examining the dataset, it was generally found to be clean and suitable for analysis. However, a few issues required attention before proceeding to the modeling stage. Some numeric fields contained zero or missing values, which might affect the model's interpretation of network behavior. In addition, a few categorical variables displayed inconsistent formatting or spacing. After addressing these issues by ensuring consistent encoding and normalization, the dataset became ready for model evaluation and machine learning experiments in WEKA.

III. DATA PREPARATION

Your task is to import the dataset into a MySQL DB table.

Reflection (40 pts):

1) Provide a detailed description of the process you performed on the dataset during the data preparation stage? Was this necessary based on your observations from the “Data Understanding” stage? (30 pts)

- During the data preparation phase, I imported the CSV file into a MySQL database to validate its structure and perform necessary preprocessing tasks. The preparation process included cleaning the dataset by removing records with missing or inconsistent values, checking data types to ensure proper numerical and categorical assignments, and standardizing categorical entries for consistency. I also normalized numerical attributes such as byte counts and duration to prevent the model from being biased toward attributes with large value ranges. Additionally, I encoded all nominal variables properly so that WEKA could interpret them during model training.

These steps were necessary based on the issues observed during the Data Understanding stage. Some fields required standardization and normalization to ensure the models could interpret all features uniformly. Proper data preparation significantly improved the performance of the classification models, as it reduced noise and eliminated inconsistencies that might have affected the accuracy of the predictive results. Overall, the preparation process ensured that the dataset was well-structured and suitable for machine learning analysis in WEKA.

2) Upload your MySQL DB table dump to your own Git branch. (10 pts)

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

```
CREATE TABLE IF NOT EXISTS `wk12_dataset_db`.`20251025__dataset__wk12` (`COL 1` varchar(16), `COL 2` varchar(17), `COL 3` varchar(22), `COL 4`  
varchar(11), `COL 5` varchar(16), `COL 6` varchar(8), `COL 7` varchar(13), `COL 8` varchar(11), `COL 9` varchar(12), `COL 10` varchar(18), `COL 11`  
varchar(14), `COL 12` varchar(15), `COL 13` varchar(11), `COL 14` varchar(16), `COL 15` varchar(12), `COL 16` varchar(14), `COL 17` varchar(25), `COL  
18` varchar(147), `COL 19` varchar(15), `COL 20` varchar(47), `COL 21` varchar(17), `COL 22` varchar(13), `COL 23` varchar(14), `COL 24` varchar(10))  
DEFAULT CHARACTER SET utf8 COLLATE utf8_general_ci;
```

[Edit inline] [Edit] [Create PHP code]

✔ 40001 rows inserted. (Query took 0.0199 seconds.)

```
INSERT INTO `wk12_dataset_db`.`20251025__dataset__wk12` (`COL 1`, `COL 2`, `COL 3`, `COL 4`, `COL 5`, `COL 6`, `COL 7`, `COL 8`, `COL 9`, `COL 10`,  
`COL 11`, `COL 12`, `COL 13`, `COL 14`, `COL 15`, `COL 16`, `COL 17`, `COL 18`, `COL 19`, `COL 20`, `COL 21`, `COL 22`, `COL 23`, `COL 24`) VALUES  
( 'Timestamp', 'Source IP Address', 'Destination IP Address', 'Source Port', 'Destination Port', 'Protocol', 'Packet Length', 'Packet Type',  
'Traffic Type', 'Malware Indicators', 'Anomaly Scores', 'Alerts Warnings', 'Attack Type', 'Attack Signature', 'Action Taken', 'Severity Level', 'User  
Information', 'Device Information', 'Network Segment', 'Geo-location Data', 'Proxy Information', 'Firewall Logs', 'IDS IPS Alerts', 'Log Source'),  
( '30/05/2023 6:33', '103.216.15.12', '84.9.164.252', '31225', '17616', 'ICMP', '503', 'Data', 'HTTP', 'IoC Detected', '28.67', '', 'Malware', 'Known
```

[Edit]

IV. MODELING

Your task is to use the three (3) newly installed classification algorithms to perform modeling and evaluate their performance.

Reflection (40 pts): 1) Using the three (3) new classification algorithms, complete the table below with your recorded measurements. (30 pts):

NAIVE BAYES

Correctly Classified Instances	40000
Incorrectly Classified Instances	0
Kappa statistics	1
Precision (Class: NO)	1.000
Precision (Class: YES)	1.000
Recall (Class: NO)	1.000
Recall (Class: YES)	1.000
F-Measures (Class: NO)	1.000
F-Measures (Class: YES)	1.000

DECISION TABLE

Correctly Classified Instances	14516
Incorrectly Classified Instances	25484
Kappa statistics	0.044
Precision (Class: NO)	0.365
Precision (Class: YES)	0.361
Recall (Class: NO)	0.351
Recall (Class: YES)	0.422
F-Measures (Class: NO)	0.358
F-Measures (Class: YES)	0.389

J48

Correctly Classified Instances	13428
Incorrectly Classified Instances	26572
Kappa statistics	0
Precision (Class: NO)	?
Precision (Class: YES)	0.336
Recall (Class: NO)	0.000
Recall (Class: YES)	1.000
F-Measures (Class: NO)	?
F-Measures (Class: YES)	0.503